

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. **Project No.:** WW-4705R
Project Name: Hilo WWTP Phase 1 **Submittal Number:**
Submittal Title: For Information Only
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authored By:	Date Submitted:

Submittal Certification	
Check Either (A) or (B): ☐ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> . ☐ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.	
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title:	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM: Date Received PM/CM to Reviewer: Date Received Reviewer to PM/CM: Date Sent PM/CM to GC:

Nan, Inc

 PROJECT: HILO WWTP REHABILITATION
 AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

 THIS SUBMITTAL HAS BEEN CHECKED BY
 THIS CONTRACTOR. IT IS CERTIFIED
 CORRECT, COMPLETE, AND IN
 COMPLIANCE WITH CONTRACT
 DRAWINGS AND SPECIFICATIONS. ALL
 AFFECTED CONTRACTORS AND
 SUPPLIERS ARE AWARE OF, AND WILL
 INTEGRATE THIS SUBMITTAL (UPON
 APPROVAL) INTO THEIR OWN WORK.

 DATE RECEIVED _____
 SPECIFICATION SECTION # _____
 SPECIFICATION _____
 PARAGRAPH _____
 DRAWING _____
 SUBCONTRACTOR _____
 SUPPLIER _____
 MANUFACTURER _____
CERTIFIED BY CQCM or Designee : 



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 16130-001.0
Date: 9/11/2025

Client: County of Hawai'i **Carollo Project No.:** 203975
Contractor: Nan, Inc.
Submittal Name: Conduits Flexible Conduit
Reviewed By: Gavin Goo

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

- No product data attached.
Product data is now attached.

Submittal



SUB Reference No : MECI-SUB-0006

Status: OUTSTANDING

For Action: Darrin Lee, MECI
David Wieseler, MECI

Project: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT

Subject: 16130 - CONDUIT - FLEXIBLE CONDUIT

Submittal for flexible conduit including sealtight-liquidtight flexible metallic conduit, liquidtight connectors, non-metallic liquidtight conduit, and explosion proof flexible conduit.

Submitted For: Approval

Specification Reference : 16130 - CONDUITS

Paragraph No.: 16130 - 1.04.B

Description : Submittal for flexible conduit including sealtight-liquidtight flexible metallic conduit, liquidtight connectors, non-metallic liquidtight conduit, and explosion proof flexible conduit.

Discipline: ELEC

Area: 00

Attachments: HWWTP - 16130 - CONDUITS - FLEXIBLE CONDUIT.pdf

SUB by:	Hannah Anderson, MECI	On: 08 September 2025
Created By:	Hannah Anderson, MECI	On: 08 September 2025, 02:22:51 PM -10:00
Last Edited By:	Hannah Anderson, MECI	On: 08 September 2025, 02:22:51 PM -10:00

01 - QC/Project Engineer Review

Approved without comment.

QC/Project Engineer Review by:	David Wieseler, MECI	On: 08 September 2025
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02 - Project Manager Review

Project Manager Review
by:

On:

Created By:On: ,

Last Edited By:On: ,

03 - Response

Response by:On:

Created By:On: ,

Last Edited By:On: ,

04 - Project Manager/Engineer Closeout

Project Manager/Engineer
Closeout by:On:

Created By:On: ,

Last Edited By:On: ,

REVISION	DATE	PREPARED BY	APPROVED BY
0	09/08/2025	Hannah Anderson	David Wieseler

HILO WASTEWATER TREATMENT PLANT REHABILITATION AND REPLACEMENT PROJECT



SPECIFICATION - 16130 FLEXIBLE CONDUIT

DOCUMENT REVISION LOG						
Revision Number	Revision Date	Description	Approvals			
			PE INITIAL		PM INITIAL	
0	09/08/2025	Issued for Review	DCW	09/08/2025		

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SECTION 16130

CONDUITS

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes:
 - 1. Metallic conduits.
 - 2. Nonmetallic conduits.
 - 3. Conduit bodies.
 - 4. Conduit fittings and accessories.
 - 5. Conduit installation.

1.02 REFERENCES

- A. Abbreviations:
 - 1. GRC: Galvanized rigid steel conduit.
 - 2. PCS: Polyvinyl chloride (PVC) coated rigid steel conduit.
 - 3. PVC: Polyvinyl chloride rigid nonmetallic conduit.
 - 4. SLT: Sealtight-liquidtight flexible conduit.
 - 5. EFLX: Explosion proof flexible conduit.
 - 6. NPT: National pipe thread.
- B. Definitions:
 - 1. Conduit bodies: A separate portion of a conduit system that provides access through a removable cover to the interior of the system at a junction of 2 or more conduit sections. Includes, but not limited to, Shapes C, E, LB, T, X, etc.
 - 2. Conduit fitting: An accessory that primarily serves a mechanical purpose. Includes, but not limited to, bushings, locknuts, hubs, couplings, reducers, etc.
- C. Standards:
 - 1. American National Standards Institute (ANSI):
 - a. C80.1 - Electrical Rigid Steel Conduit.
 - 2. National Electrical Code (NEC).
 - 3. National Electrical Manufacturer's Association (NEMA):
 - a. RN-1 - Polyvinyl Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Steel Conduit.
 - b. TC7 - Smooth-Wall Coilable Electrical Polyethylene Conduit.
 - 4. Underwriters Laboratories (UL), Inc.:
 - a. 651B - Standard for Continuous Length HDPE Conduit.
 - b. 1203 -Standard for Explosion-Proof and Dust-Ignition-Proof Electrical Equipment for Use in Hazardous (Classified) Locations.

1.03 DELEGATED DESIGN

- A. As specified in Sections 01357 - Delegated Design Procedures and 16070 - Hangers and Supports.

1.04 SUBMITTALS

- A. Furnish submittals as specified in Section 01330 - Submittal Procedures.
- B. Product data:
 - 1. Furnish complete manufacturer's catalog sheets for every type and size of conduit, fitting, conduit body, and accessories to be used on the Project.
 - 2. Furnish complete manufacturer's recommended special tools to be used for installation if required.
 - 3. Certified test results for PVC-coated metallic conduit showing the adhesive bond is stronger than the tensile strength of the PVC.
- C. Shop drawings:
 - 1. Furnish conduit routing plans for conduits before the installation of any conduit. Conduit routing submittals are to be stamped and sealed by a registered Professional Engineer. Include the following details.
 - a. Intended routing of each conduit.
 - b. Conduit size.
 - c. Conduit material.
 - d. Number and type of conductors.
 - e. Supporting methods.
 - 2. Provide a fully developed contractor conduit schedule that includes tags for all conduits and shows all wires from each source to destination.
 - a. Number conduits in accordance with the Contract Documents.
 - 3. Provide ampacity and conduit sizing calculations when combining wires within common conduits beyond or differently than what is already shown as combined in the conduit schedule. Include de-rating factors used when combining current-carrying conductors within each conduit. Match the conduit tags in the detailed contractor drawings and conduit schedule.
 - a. Reference General Notes on drawing 00-E-01-002 for constraints when combining wires within common conduits.
- D. Delegated design submittals:
 - 1. As specified in Section 16070 - Hangers and Supports.
- E. Certifications:
 - 1. Furnish PVC-coated conduit manufacturer's valid, unexpired certification for each installer.
- F. Record Documents:
 - 1. Incorporate all changes in conduit routing on electrical plan drawings.
 - 2. Dimension underground and concealed conduits from building lines.
 - 3. Furnish hard copy drawings.
- G. Installation drawings: Installation drawings, including individual conduit numbers, routing, sizes, cable sizes, and circuit numbers for each conduit.

1.05 QUALITY ASSURANCE

- A. All conduits, conduit bodies, and fittings shall be UL listed and labeled.

- B. Every installer of PVC-coated metallic conduit shall be certified by the manufacturer for installation of the conduit, and be able to present a valid, unexpired installer certification card prior to installation beginning.

1.06 DELIVERY, STORAGE, AND HANDLING

- A. Do not expose non-metallic conduit to direct sunlight.
- B. Do not store conduit in direct contact with the ground.

1.07 PROJECT OR SITE CONDITIONS

- A. As specified in Section 01850 - Design Criteria.

1.08 ADMINISTRATIVE REQUIREMENTS

- A. Sequencing:
 - 1. Before performing any trenching locate existing underground utilities:
 - a. As specified in Section 02280 – Subsurface Utility Engineering.
 - b. Review of existing civil record drawings for recorded underground utilities.
 - 1) Determine underground utility horizontal and vertical location by at least 1 of the following methods:
 - a) Soft excavation.
 - b) Local utility location service, CALL BEFORE YOU DIG or equal.

1.09 WARRANTY

- A. As specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 DESIGN AND PERFORMANCE CRITERIA

- A. Provide conduits, conduit bodies, fittings, junction boxes, and all necessary components, whether or not indicated on the Drawings, as required, to install a complete electrical raceway system.
- B. Provide location and protection of existing underground utilities, underground conduit trenching, conduit and backfill necessary for the complete installation of underground conduits.

2.02 MANUFACTURERS

- A. Galvanized rigid steel conduit:
 - 1. One of the following or equal:
 - a. Allied Tube and Conduit.
 - b. Western Tube and Conduit.
 - c. Wheatland Tube Co.

- B. PVC-coated rigid steel conduit:
 - 1. One of the following or equal:
 - a. Allied.
 - b. Calbond.
 - c. NEC, Inc. BlackGuard.
 - d. Ocal, Inc.
 - e. Robroy Ind.
- C. Sealtight-liquidtight flexible conduit:
 - 1. One of the following or equal:
 - a. Southwire.
 - b. AFC Cable Systems.
 - c. Electri-Flex Co.
 - d. Anaconda.
- D. Explosion proof flexible conduit:
 - 1. One of the following or equal:
 - a. Appleton.
 - b. Crouse-Hinds.
 - c. Hubbell Killark.
- E. Rigid nonmetallic PVC conduit:
 - 1. One of the following or equal:
 - a. Carlon.
 - b. Cantex.
 - c. Triangle Conduit and Cable.
- F. Conduit bodies:
 - 1. One of the following or equal:
 - a. Crouse-Hinds.
 - b. Appleton.
 - c. O-Z/Gedney.
 - d. Ocal, Inc.
 - e. Robroy Ind.
 - f. Calbond.
 - g. Carlon.
- G. Joint compound:
 - 1. The following or equal:
 - a. Thomas & Betts.
- H. Galvanized rigid steel conduit expansion fittings:
 - 1. One of the following or equal:
 - a. Crouse-Hinds.
 - b. Appleton.
 - c. O-Z/Gedney.
- I. PVC-coated rigid steel conduit expansion fittings:
 - 1. One of the following or equal:
 - a. Ocal, Inc.

- b. Robroy Ind.
 - c. NEC, Inc. BlackGuard.
- J. Conduit sleeve:
 - 1. One of the following or equal:
 - a. Crouse-Hinds.
 - b. Appleton.
 - c. O-Z/Gedney.
- K. Conduit seals:
 - 1. One of the following or equal:
 - a. Appleton.
 - b. Crouse-Hinds.
 - c. O-Z/Gedney.
- L. Conduit through wall and floor seals:
 - 1. The following or equal:
 - a. O-Z/Gedney:
 - 1) Type "WSK."
 - 2) Type "CSM."

2.03 COMPONENTS

- A. GRC:
 - 1. All threads: NPT standard conduit threads with a 3/4-inch taper per foot:
 - a. Running conduit threads are not acceptable.
 - 2. Hot-dip galvanized inside and out:
 - a. Ensures complete coverage and heats the zinc and steel to a temperature that ensures the zinc alloys with the steel over the entire surface.
 - b. Electro-galvanizing is not acceptable.
 - 3. Manufactured in accordance with:
 - a. UL-6.
 - b. ANSI C80.1.
- B. PCS:
 - 1. The steel conduit, before PVC coating, shall be new, unused, hot-dip galvanized material, conforming to the requirements for Type GRC.
 - 2. Coated conduit NEMA Standard RN-1:
 - a. The galvanized coating may not be disturbed or reduced in thickness during the cleaning and preparatory process.
 - 3. Factory-bonded PVC jacket:
 - a. The exterior galvanized surfaces shall be coated with primer before PVC coating to ensure a bond between the zinc substrate and the PVC coating.
 - b. Nominal thickness of the exterior PVC coating shall be 0.040 inch except where part configuration or application of the piece dictates otherwise.
 - c. PVC coating on conduits and associated fittings shall have no sags, blisters, lumps, or other surface defects and shall be free of holes and holidays.

- d. The PVC adhesive bond on conduits and fittings shall be greater than the tensile strength of the PVC plastic coating:
 - 1) Confirm bond with certified test results.
- 4. A urethane coating shall be uniformly and consistently applied to the interior of all conduits and fittings:
 - a. Nominal thickness of 0.002 inch.
 - b. Conduits having areas with thin or no coating are not acceptable.
 - c. All threads shall be coated with urethane.
- 5. The PVC exterior and urethane interior coatings applied to the conduits shall afford sufficient flexibility to permit field bending without cracking or flaking at temperature above 30 degrees Fahrenheit (-1 degree Celsius).
- 6. PCS conduit bodies and fittings:
 - a. Malleable iron.
 - b. The conduit body, before PVC coating, shall be new, unused material and shall conform to appropriate UL standards.
 - c. The PVC coating on the outside of conduit bodies shall be 0.040-inch thick and have a series of ribs to protect the coating from tool damage during installation.
 - d. 0.002-inch interior urethane coating.
 - e. Utilize the PVC coating as an integral part of the gasket design.
 - f. Stainless steel cover screw heads shall be encapsulated with plastic to ensure corrosion protection.
 - g. A PVC sleeve extending 1 conduit diameter or 2 inches, whichever is less, shall be formed at each female conduit opening.
 - 1) The inside diameter of the sleeve shall be the same as the outside diameter of the conduit to be used.
 - 2) The sleeve shall provide a vapor- and moisture resistant seal at every connection.
 - 3) Fittings shall be Form 8 and supplied with plastic encapsulated stainless steel cover screws. Fittings shall be UL Type 4X. Fittings shall be from the same manufacturer as the conduit in order to maintain system continuity and warranty.

C. SLT:

- ✓ 1. Temperature rated for use in the ambient temperature at the installed location but not less than the following:
 - ✓ a. General purpose:
 - 1) Temperature range: -20 degrees Celsius to +80 degrees Celsius.
 - ✓ b. Oil-resistant:
 - 1) Temperature range: -20 degrees Celsius to +60 degrees Celsius.
- ✓ 2. Sunlight-resistant, weatherproof, and watertight.
- ✓ 3. Manufactured from single strip steel, hot-dip galvanized on all 4 sides before conduit fabrication.
- ✓ 4. Strip steel spiral wound resulting in an interior that is smooth and clean for easy wire pulling.
- ✓ 5. Overall PVC jacket.
- ✓ 6. With integral copper ground wire, built in the core, in conduit trade sizes 1/2 inch through 1-1/4 inch.

D. EFLX:

- ✓ 1. Suitable for the hazardous Class and Group where installed:
 - ✓ a. As specified in Section 16050 - Common Work Results for Electrical.
- ✓ 2. Metallic braid shall provide continuous electrical path.
- ✓ 3. Stainless steel construction.
- ✓ 4. Provide fittings and unions as required for the installation.

E. PVC:

- 1. Extruded from virgin PVC compound:
 - a. Schedule 40 unless otherwise specified.
 - b. Schedule 80 extra-heavy wall where specified.
- 2. Rated for 90 degrees Celsius conductors or cable.
- 3. Rated for use in direct sunlight.

F. Conduit bodies:

- 1. Material consistent with conduit type:
 - a. Malleable iron bodies and covers when used with Type GRC.
 - b. PVC-coated malleable iron bodies and covers when used with Type PCS.
- 2. Conduit bodies to conform to Form 8, Mark 9, or Mogul design:
 - a. Mogul design conforming to NEC requirements for bending space for large conductors for conduit trade sizes of 1 inch and larger with conductors #4 AWG and larger, or where required for wire-bending space.
- 3. Gasketed covers attached to bodies with stainless steel screws secured to threaded holes in conduit body.

2.04 ACCESSORIES

A. Connectors and fittings:

- 1. Manufactured with compatible materials to the corresponding conduit.

B. Insulated throat metallic bushings:

- 1. Construction:
 - a. Malleable iron or zinc-plated steel when used with steel conduit.
 - b. Positive metallic conduit end stop.
 - c. Integrally molded non-combustible phenolic-insulated surfaces rated at 150 degrees Celsius.
 - d. Use fully insulated bushings on nonmetallic conduit system made of high-impact 150 degrees Celsius rated non-combustible thermosetting phenolic.

C. Insulated grounding bushings:

- 1. Construction:
 - a. Malleable iron or steel, zinc-plated, with a positive metallic end stop.
 - b. Integrally molded non-combustible phenolic-insulated surfaces rated at 150 degrees Celsius.
 - c. Tin-plated copper grounding saddle for use with copper or aluminum conductors.

D. Electrical unions (Erickson Couplings):

- 1. Construction:
 - a. Malleable iron for use with steel conduit.

- b. PVC-coated malleable iron for use with PCS conduit.
 - c. Concrete tight, 3-piece construction.
 - d. Rated for Class I Division 1 Group D in hazardous areas.
- ✓ E. SLT fittings:
 - ✓ 1. Construction:
 - ✓ a. Malleable iron.
 - ✓ b. Furnished with locknut and sealing ring.
 - ✓ c. Liquidtight, raintight, oiltight.
 - ✓ d. Insulated throat.
 - ✓ e. Furnish as straight, 45-degree elbows, and 90-degree elbows.
 - ✓ f. Designed to prevent sleeving:
 - ✓ 1) Verify complete bonding of the raceway jacket to the plastic gasket seal.
 - ✓ g. Equipped with grounding device to provide ground continuity irrespective of raceway core construction. Grounding device, if inserted into raceway and directly in contact with conductors, shall have rolled-over edges for sizes under 5 inches.
 - ✓ h. Where terminated into a threadless opening using a threaded hub fitting, a suitable moisture-resistant/oil-resistant synthetic rubber gasket shall be provided between the outside of the box or enclosure and the fitting shoulder. Gasket shall be adequately protected by and permanently bonded to a metallic retainer.
 - ✓ 2. Corrosion-resistant and outdoor SLT fittings:
 - ✓ a. Construction:
 - ✓ 1) PVC-coated liquidtight fittings with a bonded 0.040-inch-thick PVC coating on the metal connector to form a seal around the SLT conduit.
 - ✓ 2) Insulated throat and an integral sealing ring.
- F. Hubs for threaded attachment of steel conduit to sheet metal enclosures:
 - 1. Construction:
 - a. Insulated throat.
 - b. PVC-coated when used in corrosive areas.
 - c. Bonding locknut.
 - d. Recessed neoprene o-ring to ensure watertight and dusttight connector.
 - e. 1/2-inch through 1-1/4-inch steel zinc electroplated.
 - f. 1-1/2-inch through 6-inch malleable iron zinc plated.
 - 2. Usage:
 - a. All conduits in damp, wet, outdoor, and corrosive areas shall use threaded hubs for connections to sheet metal enclosures.
- G. Sealing fittings:
 - 1. Construction:
 - a. 40-percent wire fill capacity.
 - b. PVC-coated when used in corrosive areas.
 - c. PVC Coated Hazardous (Classified) Location fittings must be UL 1203 listed after the coating is applied and have a red metal tag attached to the fitting to signify compliance.
 - d. Malleable ductile iron with steel conduit.
 - e. Aluminum with aluminum conduit.

- f. Type EYDX where drains are required.
 - g. Type EYSX where drains are not required.
 - h. UL 1203 listed for use in Class I, Division 1, Groups A, B, C, D; Class I, Division 2, Groups A, B, C, D; and Class II, Divisions 1 and 2, Groups E, F, and G.
- 2. Sealing compound:
 - a. Fiber filler and cement as recommended by the sealing fitting manufacturer.
 - b. Approved for the conditions and use.
 - 1) Not affected by surrounding atmosphere or liquids.
 - c. Melting point shall be 200 degrees Fahrenheit minimum.
- H. Expansion/deflection couplings:
 - 1. Use to compensate for movement in any directions between 2 conduit ends where they connect.
 - 2. Shall allow movement of 3/4 inch from the normal in all directions.
 - 3. Shall allow angular movement for a deflection of 30 degrees from normal in any direction.
 - 4. Constructed to maintain electrical continuity of the conduit system.
 - 5. Materials:
 - a. End couplings: Bronze or galvanized ductile iron.
 - a. Sleeve: Neoprene.
 - b. Bands: Stainless steel.
 - c. Bonding jumper: Tinned copper braid.
- I. Expansion couplings:
 - 1. Shall allow for expansion and contraction of conduit:
 - a. Permitting 8-inch movement, 4 inches in either direction.
 - 2. Constructed to maintain electrical continuity of the conduit system.
 - 3. Materials:
 - a. Head: Malleable or ductile iron.
 - b. Sleeve: Steel.
 - c. Insulating bushing: Phenolic.
 - d. Finish: Hot-dip galvanized.
 - e. PVC-coated steel when used with Type PCS.
- J. Conduit markers:
 - 1. As specified in Section 16075 - Identification for Electrical Systems.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Before installing any conduit or locating any device box:
 - 1. Examine the complete set of Drawings and Specifications, and all applicable shop drawings.
- B. Verify all dimensions and space requirements and make any minor adjustments to the conduit system as required to avoid conflicts with the building structure, other equipment, or the work of other trades.

3.02 INSTALLATION

A. General:

1. Conduit routing:
 - a. The electrical drawings are diagrammatic in nature. Conduit routing is indicated on the drawings using both "homeruns" and schematic representation.
 - 1) Install conduit homeruns as indicated on the Drawings and as specified.
 - 2) Install conduit runs with schematic representation indicated on the Drawings and as specified.
 - 3) Modify conduit runs to suit field conditions, as accepted by the Engineer:
 - a) Make changes in conduit locations that are consistent with the design intent but are dimensionally different, or routing to bypass obstructions.
 - b) Make changes in conduit routing due to the relocation of equipment.
 - c) Install conduits and equipment in such a manner as to avoid obstructions and to preserve headroom and keep openings and passageways clear.
 - 4) Where the Drawings do not indicate the exact mounting and/or supporting method to be used, use materials and methods similar to the mounting details indicated on the Drawings.
 - 5) The electrical drawings do not indicate all required junction boxes and pull boxes:
 - a) Provide junction boxes and pull boxes to facilitate wire pulling as required:
 - (1) To meet cable manufacturer's pulling tension requirements.
 - (2) To limit total conduit bends between pull locations.
 - b) Install junction boxes and pull boxes at locations acceptable to the Engineer.
 - b. The Contractor is responsible for any deviations in general location, conduit size, routing, or changes to the conduit schedule without the express written approval or direction by the Engineer:
 - 1) The Engineer is the sole source in determining whether the change is constituted as a deviation:
 - a) Perform any changes resulting in additional conduits, or extra work from such deviations.
 - b) Incorporate any deviations on the Record Documents.
 - 2) Owner reserves the right to deduct the amount of applicable reimbursement, equivalent to the cost of the engineering effort required to show those unauthorized changes on As-Built Drawings.
2. Use only tools recommended by the conduit manufacturer for assembling the conduit system.
3. Provide adequate clearances from high-temperature surfaces for all conduit runs. Provide minimum clearances as follows:
 - a. Clearance of 6 inches from surfaces 113 degrees Fahrenheit to 149 degrees Fahrenheit.
 - b. Clearance of 12 inches from surfaces greater than 149 degrees Fahrenheit.

- c. Keep conduits at least 6 inches from the coverings on hot water and steam pipes, 18 inches from the coverings on flues and breechings, and 12 inches from fuel lines and gas lines.
 - d. Where it is necessary to route conduits close to high-temperature surfaces, provide a high-reflectance thermal barrier between the conduit and the surface.
- 4. Support conduit runs on water-bearing walls a minimum of 7/8-inch away from wall on an accepted preformed channel:
 - a. Do not run conduits within water-bearing walls unless otherwise indicated on the Drawings.
- 5. Do not install 1-inch or larger conduits in or through structural members unless approved by the Engineer.
- 6. Run conduits exposed to view parallel with or at right angles to structural members, walls, or lines of the building:
 - a. Install straight and true conduit runs with uniform and symmetrical elbows, offsets, and bends.
 - b. Make changes in direction with long radius bends or with conduit bodies.
- 7. Install conduits with total conduit bends between pull locations less than or equal to 270 degrees.
- 8. Route all exposed conduits to preserve headroom, access space and workspace, and to prevent tripping hazards and clearance problems:
 - a. Install conduit runs so that runs do not interfere with proper and safe operation of equipment and do not block or interfere with ingress or egress, including equipment-removal hatches.
 - b. Route conduits to avoid drains or other gravity lines. Where conflicts occur, relocate the conduit as required.
- 9. Conduits may be run in concrete members or slabs with permission of the Engineer or as indicated on the Drawings:
 - a. Refer to the typical details for conduit spacing and size requirements.
- 10. When installing conduits through existing slabs or walls, make provisions for locating any possible conflicting items where the conduit is to penetrate. Use tone signal or X-ray methods to make certain that no penetrations will be made into the existing conduits, piping, cables, post-tensioning cables, etc.
- 11. Plug conduits brought into pull boxes, manholes, handholes, and other openings until used to prevent entrance of moisture.
- 12. Install conduits through wall and floor seals where indicated on the Drawings.
- 13. For existing and new 2-inch and larger conduit runs, snake conduits with a conduit cleaner equipped with a cylindrical mandrel of a diameter not less than 85 percent of nominal diameter of the conduit:
 - a. Remove and replace conduits through which mandrel will not pass.
- 14. Provide all sleeves and openings required for the passage of electrical raceways or cables even when these openings or sleeves are not specifically indicated on the Drawings.
- 15. Install complete conduit systems before conductors are installed.
- 16. Provide metallic conduits terminating in transformer, switchgear, motor control center, or other equipment conduit windows with grounding bushings and ground with a minimum No. 6 AWG ground wire.
- 17. Underground conduits:
 - a. Install underground conduits, including conduit runs below slabs-on-grade in concrete-reinforced duct bank construction:
 - 1) As specified in Section 16133 - Duct Banks.

- b. Make underground conduit size transitions at handholes and manholes.
 - c. Install spare conduits in underground duct banks towards top center of runs to allow for ease of installation of future cables as conduits enter underground manholes and handholes.
 - d. Seal around conduit penetrations of below grade walls with a mechanical seal.
- 18. Underground conduit trenching:
 - a. Perform trenching as specified in Section 02318 - Trenching.
 - b. Trench must be uniformly graded with the bottom, rock free and covered with select material.
 - c. Damage occurring to existing ducts, conduits, cables, and other utilities during underground conduit installation shall be remediated to the satisfaction of the Owner.
 - d. Whenever possible, use the walls of the trench as forms for concrete encasement:
 - 1) Forms are required where the soil is not self-supporting.
- B. Equipment grounding conductors:
 - 1. Provide a separate, green insulated, grounding conductor in each raceway independent of raceway material:
 - a. Multi-conductor power and control cables shall include an integral green insulated grounding conductor.
 - b. Provide a separate grounding conductor in each individual raceway for parallel feeders.
 - 2. Conductors shall be the same type and insulation as the circuit conductors:
 - a. Use 600-volt insulation for the equipment grounding conductors for medium voltage systems.
 - 3. Minimum size in accordance with the NEC.
- C. Lighting and receptacle conduits:
 - 1. Provide conduit runs for lighting and receptacle circuits, whether or not indicated on the Drawings:
 - 2. Install conduits in accordance with the requirements of this Section unless otherwise indicated.
 - 3. Minimum conduit size:
 - a. 3/4-inch for exposed conduits.
 - b. 1-inch for underground or in-slab conduits.
 - 4. Provide conduit materials for the installed location as specified in Section 16050 - Common Work Results for Electrical.
- D. Hazardous areas:
 - 1. As specified in Section 16050 - Common Work Results for Electrical for hazardous areas and specific Class and Division.
- E. Conduit usage:
 - 1. Exposed conduits:
 - a. Rigid conduit:
 - 1) Install the rigid conduit type for each location as specified in Section 16050 - Common Work Results for Electrical and 16052 – Hazardous Classified Area Construction.
 - 2) Minimum size: 3/4-inch.

- b. Flexible conduit:
- 1) Use flexible conduit for final connections between rigid conduit and motors, vibrating equipment, instruments, control equipment, or where required for equipment servicing:
 - a) Use Type SLT with rigid metallic conduit.
 - b) Use Type NFC with PVC conduit.
 - c) Use Type EFLX in Class I Division 1 locations.
 - 2) Minimum size: 3/4-inch:
 - a) 1/2 when required for connection to instruments.
 - 3) Maximum length:
 - a) Fixed equipment:

Conduit Trade Size	Flexible Conduit Length (inch)
3/4	18
1	18
1-1/4	18
1-1/2	18
2	36
2-1/2	36
3	36
3-1/2	38
4	40

- b) Removable instruments or hinged equipment:
 - (1) As required to allow complete removal or full movement without disconnecting or stressing the conduit.
2. Concrete-encased and embedded conduits:
 - a. Straight runs and bends less than 45 degrees:
 - 1) Type PVC Schedule 40.
 - b. Bends with total deflection greater than 45 degrees:
 - 1) PCS.
 - c. Entering and exiting duct bank, underground or embedded conduit runs a minimum 12 inches above and below grade, finished floor, or entering equipment:
 - 1) PCS.
 - d. Minimum size:
 - 1) 2-inch in duct banks.
 - 2) 1-inch for in-slab conduits.
 - 3) Provide conduit fittings to enlarge the conduit from the exposed size in the conduit schedule as required.
3. Direct-buried and sand-bedded duct bank conduits:
 - a. Type PCS.
 - b. Minimum size: 1-inch.
4. Below-slab conduits:
 - a. Type PCS.

- b. Minimum size: 1-inch.
 - 5. Concrete capped, pea gravel-bedded duct bank conduits:
 - a. Type PVC40.
 - b. Minimum size: 1-inch.
 - 6. PVC-coated rigid metallic conduit:
 - a. Use specifically manufactured or machined threading dies to manufacturer's specifications to accommodate the PVC jacket.
 - b. Repair damage to PVC coatings with manufacturer supplied touchup compound or PVC Coating Repair Kit for PVC Coated Raceway Systems.
 - 7. GRC:
 - a. Conduit shall be cut square and reamed before threading.
- F. Conduit joints and bends:
- 1. General:
 - a. Where conduit is underground, under slabs on grade, exposed to the weather, or in NEMA Type 4 or NEMA Type 4X locations, make joints liquidtight.
 - b. Keep bends and offsets in conduit runs to an absolute minimum.
 - c. All bends shall be symmetrical.
 - d. The following conduit systems shall use large-radius sweep elbows:
 - 1) Underground conduits.
 - 2) Conduits containing fiber optic cables.
 - e. Provide large-radius factory-made bends for 1-1/4-inch trade size or larger.
 - f. Make field bends with a radius of not less than the requirements found in the NEC:
 - 1) The minimum bending radius of the cable must be less than the radius of the conduit bend.
 - 2) Make all field bends with power bending equipment or manual benders specifically intended for the purpose:
 - a) Make bends so that the conduit is not damaged and the internal diameter is not effectively reduced.
 - b) For the serving utilities, make bends to meet their requirements.
 - g. Replace all deformed, flattened, or kinked conduit.
 - 2. Threaded conduit:
 - a. Cut threads on rigid metallic conduit with a standard conduit-cutting die that provides a 3/4-inch per foot taper and to a length such that all bare metal exposed by the threading operation is completely covered by the couplings or fittings used. In addition, cut the lengths of the thread such that all joints become secure and wrench-tight just preceding the point where the conduit ends would butt together in couplings or where conduit ends would butt into the ends or shoulders of other fittings.
 - b. Thoroughly ream conduit after threads have been cut to remove burrs.
 - c. Use bushings or conduit fittings at conduit terminations.
 - d. On exposed conduits, repair scratches and other defects with galvanizing repair stick, Enterprise Galvanizing "Galvabar™," or CRC "Zinc It."
 - e. Coat conduit threads with an approved electrically conductive sealant and corrosion inhibitor that is not harmful to the conductor insulation:
 - 1) Apply to the male threads and tighten joints securely.
 - 2) Clean excess sealant from exposed threads after assembly.
 - f. Securely tighten all threaded connections.

- g. Any exposed threaded surfaces must be cleaned and coated with a galvanizing solution so that all exposed surfaces have a galvanized protective coating.
- 3. PVC:
 - a. Use approved solvent-weld cement specifically manufactured for the purpose. Spray-type cement is not allowed.
 - b. Apply heat for bends so that conduit does not distort or discolor. Use a spring mandrel as required to ensure full inside diameter at all bends:
 - 1) Utilize a heater specifically for PVC conduit as recommended by the conduit manufacturer.
- G. Conduit sealing and drainage:
 - 1. Conduit drainage and sealing other than required for hazardous and classified areas:
 - a. Provide sealing and drainage in vertical drops of long (in excess of 20 feet), exterior, above-grade conduit runs at the points at which the conduit enters buildings, switchgear, control panels, lighting panelboards, and other similar enclosures.
 - b. Provide seal fittings with drains in vertical drops directly above grade for exterior and above-grade conduit runs that are extended below grade.
 - c. Provide conduit seals with drains in areas of high humidity and rapidly changing temperatures:
 - 1) Where portions of an interior raceway pass through walls, ceilings, or floors that separate adjacent areas having widely different temperatures.
 - d. Provide conduit seals similar to O-Z/Gedney (Type CSM) on all conduits between corrosive and non-corrosive areas.
 - e. Seal one end only of all underground conduits at highest point with O-Z/Gedney sealing (non-hazardous) filling, or equal.
 - 2. Install seals with drains at any location along conduit runs where moisture may condense or accumulate. This requirement includes, but is not limited to, the following locations: control panels, junction boxes, pullboxes, or low points of the conduit.
- H. Hangers and supports:
 - 1. General:
 - a. Provide appropriate hangers, supports, fasteners, and seismic restraints to suit applications:
 - 1) As specified in Section 16070 - Hangers and Supports.
 - 2) Provide support materials consistent with the type of conduit being installed as specified in Section 16050 - Common Work Results for Electrical.
 - b. Support conduit at the intervals required by the NEC.
 - c. Perforated strap and plumbers' tape are not acceptable for conduit supports.
 - 2. Finished areas:
 - a. Above suspended ceilings:
 - 1) Support conduit on or from the structure. Do not support conduit from hanging wires or suspended ceiling grid.

- b. Concealed conduit on wood:
 - 1) Use 2-hole galvanized steel straps screwed or nailed to the wood or hammer-driven stamped galvanized-type supports having serrated or sawtooth edges on the driven portion and designed specifically for the size and type of conduit being supported. Drive these latter supports so that the conduit is tightly and rigidly supported. Replace any dented or damaged conduit.
 - c. In steel-stud construction:
 - 1) Tie conduit at maximum 4-foot intervals with No. 16 gauge double-annealed galvanized wire or conduit clips so that conduit cannot move from vibration or other causes.
 - 3. Conduit on concrete or masonry:
 - a. Use 1-hole malleable iron straps with metallic or plastic expansion anchors and screws or support from preset inserts.
 - b. Use preset inserts in concrete when possible.
 - c. Use pipe spacers (clamp backs) in wet locations.
 - 4. Conduit on metal decking:
 - a. Use 1-hole malleable iron straps with 1-inch-long cadmium-plated Type A panhead sheet-metal screws. Fully or partially hammer-driven screws are not acceptable.
 - 5. Suspended conduit:
 - a. Use malleable-iron factory-made split-hinged pipe rings with threaded suspension rods sized for the weight to be carried (minimum 3/8-inch diameter), Kindorf, or equal.
 - b. For grouped conduits, construct racks with threaded rods and tiered angle iron or preformed channel cross members. Clamp each conduit individually to a cross member. Where rods are more than 2-feet long, provide rigid sway bracing.
 - 6. Supports at structural steel members:
 - a. Use beam clamps.
 - b. Drilling or welding may be used only as specified or with approval of the Engineer.
 - 7. PVC-coated rigid metal systems:
 - a. Provide right-angle beam clamps and "U" bolts specially formed and sized to snugly fit the outside diameter of the coated conduit. Provide "U" bolts with PVC-encapsulated nuts that cover the exposed portions of the threads.
 - b. Securely fasten exposed conduits with Type 316 stainless steel clamps or straps.
- I. Expansion or expansion/deflection fittings:
 - 1. General:
 - a. Align expansion coupling with the conduit run to prevent binding.
 - b. Follow manufacturer's instructions to set the piston opening.
 - c. Install expansion fittings across concrete expansion joints and at other locations where necessary to compensate for thermal or mechanical expansion and contraction.
 - d. Furnish fittings of the same material as the conduit system.

2. For metallic conduit, provide expansion or expansion/deflection couplings, as appropriate, where:
 - a. Install expansion fittings a minimum of every 200 feet in straight conduit runs.
- J. Empty conduits:
 1. Provide a pull tape in each empty conduit more than 10 feet in length.
 2. Seal ends of all conduits with approved, manufactured conduit seals, caps, or plugs immediately after installation:
 - a. Keep ends sealed until immediately before pulling conductors.
- K. Miscellaneous:
 1. Seal roof penetrations for raceways and other items that penetrate the roof in accordance with roofing manufacturer's instructions and as indicated on the Drawings.
 2. Provide electrical unions at all points of union between ends of rigid conduit systems that cannot otherwise be coupled:
 - a. Running threads and threadless couplings are not allowed.
 3. Replace any conduits installed that the Engineer determines do not meet the requirements of this Specification.
 4. Provide conduit housekeeping curb around all embedded or below-grade conduits exiting or entering the slab, per the Typical Details.

3.03 COMMISSIONING

- A. As specified in Section 01756 - Commissioning.

END OF SECTION

1.0 SEALTIGHT-LIQUIDTIGHT FLEXIBLE CONDUIT

SPECIFICATION – 16130-2.02.C FLEXIBLE CONDUIT



Titan®2

Type UL Liquidtight Flexible Metal Conduit



Liquidtight Flexible Metal Conduit Metal Conduit. UL Listed. CSA Listed. Oil-Resistant. Sunlight-Resistant. Temperature Rated -55°C to 80°

APPLICATIONS

Titan®2 Type UL Liquidtight Flexible Metal Conduit is suitable of the following installations:

- Used in industrial and commercial applications for conveyors, blowers, cranes, air conditioners, machine tooling and lubrication equipment
- Where the conditions of installation, operation, or maintenance require flexibility or protection from liquids, vapors, solids, or weather
- Applications requiring movement, crossover connections, or tight bends
- Exposed or concealed locations
- For flexible connections to swimming pool, spa, and hot tub motors per NEC 680.21(A)(3) & 680.42(A)(1)
- Electric signs and outline lighting supply and secondary-circuit per NEC 600.31 (1000 Volts or less) & NEC 600.32 (over 1000 Volts)
- Floating building feeders and services per NEC 553.7(B)
- Boatyards & Marinas in accordance with NEC 555.13
- Cranes & Hoists in accordance with NEC 610.11(C)
- For Elevator, Dumbwaiters, Escalators, Moving Walks, Wheel Chair Lifts & Stairway Chair Lifts in accordance with NEC 620.21 (where expressly permitted)
- Under raised floors in data processing areas per NEC 645.5(E)(2)
- Service entrance in lengths up to six feet per NEC 230.43(15)
- Permitted for equipment grounding under the conditions detailed in NEC 250.118(6)
- In Hazardous Locations - where necessary for flexible connections within hazardous locations in accordance with the following:
 - Class I, Div. 2 - NEC 501.10(B)(2) & 501.30(B)
 - Class II, Div. 1 - NEC 502.10(A)(2) & 502.30(B)
 - Class II, Div. 2 - NEC 502.10(B)(2)
 - Class III, Div. 1 - NEC 503.10(A)(3) & 503.30(B)
 - Class III, Div. 2 - NEC 503.10(B)

STANDARDS & REFERENCES

- NFPA-70 (National Electrical Code), Article 350 - Liquidtight Flexible Metal Conduit
- UL Listed to Underwriters Laboratories Standard ANSI / UL-360 for Liquidtight Flexible Steel Conduit
- CSA Listed to CSA 22.2 No.56 for use per the Canadian Electrical Code C22.1 Section 12-1300
- REACH/RoHS-2 (Chemical Limit) Compliant



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One Southwire Drive, Carrollton, GA 30119, USA



Southwire®

CONSTRUCTION

Titan®2 Type UL LFMC is manufactured with a spiral wound strip of heavy gauge, corrosion-resistant, hot-dipped galvanized steel. For 3/8" through 1-1/4" trade sizes, the core is constructed with a square locked steel strip with an integral copper-bonding strip enclosed within the steel convolutions. For 1-1/2" through 4" trade sizes, the core is constructed with a fully interlocked steel strip. A rugged, flame retardant, flexible PVC jacket is extruded over the steel core. The grey jacket resists oils, mild acids and exposure to sunlight. Also available in other colors subject to minimum runs.

TRADE SIZE (INCHES)	STOCK NUMBER			DIAMETER (INCHES)		WEIGHT (LBS/100')	APPROX. BEND DIAMETER (INCHES)
				INNER (MIN/MAX)	OUTER (MIN/MAX)		
GRAY JACKET							
3/8	55-09-29-03 (100')	55-09-29-01 (500')	55-09-29-05 (600')	.484/.504	.69/.71	24	4
1/2	55-08-26-03 (100')	55-08-26-05 (500')	55-08-26-02 (1000')	.622/.642	.82/.84	30	6.5
3/4	55-08-27-03 (100')	55-08-27-05 (500')	55-08-27-02 (1000')	.82/.84	1.03/1.05	38	8.5
1	55-09-30-03 (100')	55-09-30-04 (400')		1.041/1.066	1.29/1.315	60	13
1-1/4	55-09-33-05 (50')	55-09-33-01 (200')	55-09-33-03 (250')	1.38/1.41	1.63/1.66	80	16
1-1/2	55-09-32-05 (50')	55-09-32-01 (150')		1.575/1.6	1.865/1.90	113	18
2	55-09-31-05 (50')	55-09-31-01 (100')		2.02/2.045	2.34/2.375	149	22.25
2-1/2	55-10-64-01 (25')	55-10-64-03 (100')		2.48/2.505	2.84/2.875	197	29.25
3	55-10-65-04 (25')	55-10-65-03 (100')		3.07/3.1	3.46/3.5	265	35
3-1/2	55-10-66-04 (25')	55-10-66-03 (100')		3.5/3.54	3.96/4	300	40
4	55-10-67-04 (25')	55-10-67-02 (100')		4/4.04	4.46/4.5	310	48
BLACK JACKET							
3/8	55-29-08-04 (100')	55-29-08-05 (600')		.484/.504	.69/.71	24	4
1/2	55-28-94-04 (100')	55-28-94-21 (500')	55-28-91-01 (1000')	.622/.642	.82/.84	30	6.5
3/4	55-28-96-04 (100')	55-28-96-02 (500')	55-28-96-01 (1000')	.82/.84	1.03/1.05	38	8.5
1	55-28-98-01 (100')	55-28-98-04 (400')		1.041/1.066	1.29/1.315	60	13
1-1/4	55-29-04-02 (50')	55-29-04-21 (250')		1.38/1.41	1.63/1.66	80	16
1-1/2	55-31-62-03 (50')	55-31-62-01 (150')		1.575/1.6	1.865/1.99	113	18
2	55-29-06-02 (50')	55-29-06-01 (100')		2.02/2.045	2.34/2.375	149	22.25
3	55-37-37-04 (25')			3.07/3.1	3.46/3.5	265	35
4	55-29-05-01 (25')			4/4.04	4.46/4.5	310	48
* Minimum bend radius based on NEC Chapter 9, Table 2 (other bends) per Article 350.							



2.03.C.2

FEATURES

- A protective thermoplastic outer jacket which seals out water, liquids, abrasives, alcohol, coolants, corrosive fumes and gases, dirt, grease, mineral acids, non-concentrated fixed alkalines, petroleum oils, salt air and spray, and weather
- Smooth metal interior for easy wiring pulling
- UV sunlight resistant jacket
- Rated for temperature range of -55°C to 80°C, 60°C Oil (-67°F to +176°F, 140°F Oil)
- Accepts standard metallic liquidtight fittings
- Rated for direct burial applications including concrete encasement

2.03.C.4

2.03.C.2

2.03.C.1

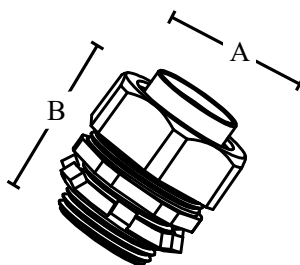


STRAIGHT LIQUIDTIGHT CONNECTORS

WITH INSULATED THROAT - MALLEABLE IRON



470S



PART #	SIZE	INNER QTY.	MASTER QTY.	WEIGHT/ CASE (LBS.)	A	B
470S	3/8" 1/2" K0	20	200	31	1.18"	1.72"
471S	1/2"	20	200	33	1.36"	1.73"
472S	3/4"	10	100	25	1.61"	1.81"
473S	1"	5	50	18	1.89"	1.85"
474S	1-1/4"	5	50	28	2.39"	2.05"
475S	1-1/2"	2	20	15	2.62"	2.38"
476S	2"	2	20	22	3.24"	2.49"
477S	2-1/2"	---	1	2	3.89"	3.21"
478S	3"	---	1	3	4.74"	3.37"
479S	3-1/2"	---	1	4	5.31"	3.40"
480S	4"	---	1	5	5.82"	3.66"
480SA*	5"	---	1	10	6.06"	4.63"
480SB*	6"	---	1	13	7.75"	4.94"

*480SA and 480SB not UL listed

FEATURES

- Liquidtight, raintight, oiltight
- Flared threaded male hub NPT
- Insulated throat protects conductors from abrasion damage
- Compact slim diameter with small turning radius
- Zinc electro plated for excellent durability

APPLICATIONS

- To secure, ground and seal liquidtight conduit to a box or enclosure
- UL listed liquidtight flexible metal conduit fittings are suitable for use in the following hazardous locations under: NEC 501-4(b), 502-4(a), (2)503-3(a) Class 1 Div.2, Class II Div.1 and 2, Class III Div.1 and 2 and are suitable for grounding in sizes 3/8" through 1-1/2" under NEC 351-1

STANDARD MATERIAL

- Malleable iron
- Steel locknut

STANDARD FINISH

- Zinc plated

CERTIFICATIONS

- UL Standard: UL 514B
- Fed. Spec: W-F-406E, W-F-406D
- NEMA: FB 1-2014

STRAIGHT LIQUIDTIGHT CONNECTORS

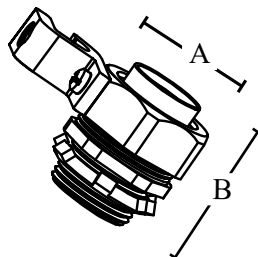
WITH GROUNDING LUG - MALLEABLE IRON



471SGR



473CSGR



FEATURES

- Liquidtight, raintight, oiltight
- Aluminum or Copper Lay-in grounding lug
- Insulated throat protects conductors from abrasion damage
- Zinc electro plated for excellent durability

APPLICATIONS

- Use to connect liquidtight conduit and terminate an external bonding conductor that runs parallel to the conduit
- The NEC code specifies length limitations and that the bonding jumper must be parallel to the conduit
- UL listed liquidtight flexible metal conduit fittings are suitable for use in the following hazardous locations under: NEC 501-4(b), 502-4(a), (2)503-3(a) Class 1 Div.2, Class II Div.1 and 2, Class III Div.1 and 2 and are suitable for grounding in sizes 3/8" through 1-1/2" under NEC 351-1

STANDARD MATERIAL

- Malleable iron
- Steel locknut
- Aluminum or Copper lug

STANDARD FINISH

- Zinc plated

CERTIFICATIONS

- UL 467 - UL 486A
- Fed. Spec: W-S-610E
- NEMA: GR 1-2017

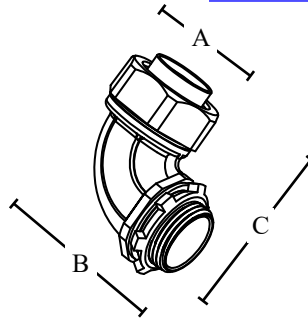
PART #	SIZE	LUG SIZE	INNER QTY.	MASTER QTY.	WEIGHT/ CASE (LBS.)	A	B
ALUMINUM GROUNDING LUG							
470SGR	3/8"	14-4	10	100	22	1.18"	1.72"
471SGR	1/2"	14-4	10	100	23	1.36"	1.73"
472SGR	3/4"	14-4	5	50	18	1.61"	1.81"
473SGR	1"	14-4	5	50	24	1.89"	1.85"
474SGR	1-1/4"	14-4	2	20	16	2.39"	2.05"
475SGR	1-1/2"	14-4	2	20	18	2.62"	2.38"
476SGR	2"	14-4	2	20	27	3.24"	2.49"
477SGR	2-1/2"	1/0-14	---	1	7	3.89"	3.21"
478SGR	3"	1/0-14	---	1	8	4.74"	3.37"
479SGR	3-1/2"	6/250	---	1	8	5.31"	3.40"
480SGR	4"	6/250	---	1	10	5.82"	3.66"
COPPER GROUNDING LUG							
471CSGR	1/2"	14-4	10	100	24	1.36"	1.73"
472CSGR	3/4"	14-4	5	50	16	1.61"	1.81"
473CSGR	1"	14-4	5	50	22	1.89"	1.85"
474CSGR	1-1/4"	14-4	2	20	14	2.39"	2.05"
475CSGR	1-1/2"	14-4	2	20	17	2.62"	2.38"
476CSGR	2"	14-4	2	20	26	3.24"	2.49"
MSTR-250-CL	2-1/2"	1/0-14	---	1	7	3.89"	3.21"
MSTR-300-CL	3"	1/0-14	---	1	8	4.74"	3.37"
MSTR-350-CL	3-1/2"	6/250	---	1	8	5.31"	3.40"
MSTR-400-CL	4"	6/250	---	1	10	5.82"	3.66"

90° LIQUIDTIGHT CONNECTORS

WITH INSULATED THROAT - MALLEABLE IRON



490S



2.04.E.1.d

2.04.E.1.c

FEATURES

- Liquidtight, raintight, oiltight
- Suitable for wet location use
- Tapered threaded male hub NPT
- Insulated throat protects conductors from abrasion damage
- Compact slim diameter with small turning radius
- Zinc electro plated for excellent durability

APPLICATIONS

- To secure, ground and seal liquidtight conduit to a box or enclosure
- UL listed liquidtight flexible metal conduit fittings are suitable for use in the following hazardous locations under: NEC 501-4(b), 502-4(a), (2)503-3(a) Class 1 Div.2, Class II Div.1 and 2, Class III Div.1 and 2 and are suitable for grounding in sizes 3/8" through 1-1/2" under NEC 351-1

STANDARD MATERIAL

- Malleable iron
- Steel locknut

2.04.E.1.a

2.04.E.1.b

STANDARD FINISH

- Zinc plated



PART #	SIZE	INNER QTY.	MASTER QTY.	WEIGHT/ CASE (LBS.)	A	B	C
490S	3/8" 1/2" KO	10	100	26	1.18"	1.96"	2.47"
491S	1/2"	10	100	28	1.36"	2.10"	2.54"
492S	3/4"	10	100	41	1.61"	2.45"	2.82"
493S	1"	5	50	31	1.89"	2.83"	3.08"
494S	1-1/4"	2	20	20	2.39"	3.15"	3.53"
495S	1-1/2"	2	20	25	2.62"	3.50"	4.17"
496S	2"	1	10	19	3.24"	4.02"	4.69"
497S	2-1/2"	---	1	5	3.89"	6.23"	6.74"
498S	3"	---	1	7	4.74"	6.91"	7.52"
499S	3-1/2"	---	1	9	5.31"	7.57"	7.93"
500S	4"	---	1	14	5.82"	8.07"	8.72"

90° LIQUIDTIGHT CONNECTORS

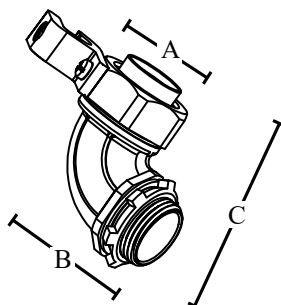
WITH GROUNDING LUG - MALLEABLE IRON



490SGR



490CSGR



FEATURES

- Liquidtight, raintight, oiltight
- Aluminum or Copper Lay-in grounding lug
- Insulated throat protects conductors from abrasion damage
- Zinc electro plated for excellent durability

APPLICATIONS

- Use to connect liquidtight conduit and terminate an external bonding conductor that runs parallel to the conduit
- The NEC code specifies length limitations and that the bonding jumper must be parallel to the conduit
- UL listed liquidtight flexible metal conduit fittings are suitable for use in the following hazardous locations under: NEC 501-4(b), 502-4(a), (2)503-3(a) Class 1 Div.2, Class II Div.1 and 2, Class III Div.1 and 2 and are suitable for grounding in sizes 3/8" through 1-1/2" under NEC 351-1

STANDARD MATERIAL

- Malleable iron
- Steel locknut
- Aluminum or Copper lug

STANDARD FINISH

- Zinc plated

CERTIFICATIONS

- UL 467 - UL 486A
- Fed. Spec: W-S-610E
- NEMA: GR 1-2017



PART #	SIZE	LUG SIZE	INNER QTY.	MASTER QTY.	WEIGHT/ CASE (LBS.)	A	B	C
ALUMINUM GROUNDING LUG								
490SGR	3/8"	14-4	5	50	17	1.18"	1.96"	2.47"
491SGR	1/2"	14-4	5	50	24	1.36"	2.10"	2.54"
492SGR	3/4"	14-4	2	20	16	1.61"	2.45"	2.82"
493SGR	1"	14-4	2	20	17	1.89"	2.83"	3.08"
494SGR	1-1/4"	14-4	2	20	25	2.39"	3.15"	3.53"
495SGR	1-1/2"	14-4	2	20	29	2.62"	3.50"	4.17"
496SGR	2"	14-4	2	20	42	3.24"	4.02"	4.69"
497SGR	2-1/2"	1/0-14	---	1	9	3.89"	6.23"	6.74"
498SGR	3"	1/0-14	---	1	11	4.74"	6.91"	7.52"
499SGR	3-1/2"	6/250	---	1	13	5.31"	7.57"	7.93"
500SGR	4"	6/250	---	1	15	5.82"	8.07"	8.72"
COPPER GROUNDING LUG								
491CSGR	1/2"	14-4	5	50	18	1.36"	2.10"	2.54"
492CSGR	3/4"	14-4	2	20	11	1.61"	2.45"	2.82"
493CSGR	1"	14-4	2	20	15	1.89"	2.83"	3.08"
494CSGR	1-1/4"	14-4	2	20	21	2.39"	3.15"	3.53"
495CSGR	1-1/2"	14-4	2	20	26	2.62"	3.50"	4.17"
496CSGR	2"	14-4	2	20	41	3.24"	4.02"	4.69"

90° LIQUIDTIGHT CONNECTORS

304 STAINLESS STEEL



LTCSS-5090



PART #	SIZE	MASTER QTY.
LTCSS-5090	1/2"	20
LTCSS-7590	3/4"	20
LTCSS-10090	1"	5
LTCSS-12590	1-1/4"	5
LTCSS-15090	1-1/2"	5
LTCSS-20090	2"	5

FEATURES

- For use with metallic and non-metallic type B
- Connectors can be easily disassembled and reused
- Sealing washers included
- Heavy duty stainless steel lock nuts included

APPLICATIONS

- Connects liquid-tight conduit to a box or enclosure in wet, damp & corrosive locations

STANDARD MATERIAL

- 304 stainless steel

CERTIFICATIONS

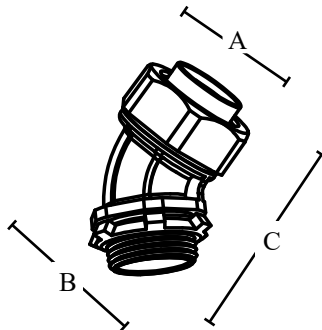
- UL Listed, CSA Listed

45° LIQUIDTIGHT CONNECTORS

WITH INSULATED THROAT - MALLEABLE IRON



230S



PART #	SIZE	INNER QTY.	MASTER QTY.	WEIGHT/ CASE (LBS.)	A	B	C
230S	3/8" 1/2" KO	10	100	24	1.18"	1.39"	2.51"
231S	1/2"	10	100	28	1.36"	1.50"	2.51"
232S	3/4"	10	100	42	1.61"	1.73"	2.70"
233S	1"	5	50	32	1.89"	2.11"	3.05"
234S	1-1/4"	2	20	22	2.39"	2.56"	3.65"
235S	1-1/2"	2	20	32	2.62"	2.77"	4.01"
236S	2"	2	20	49	3.24"	3.38"	4.64"
237S	2-1/2"	---	1	5	3.89"	4.68"	6.46"
238S	3"	---	1	7	4.74"	5.33"	7.01"
239S	3-1/2"	---	1	10	5.31"	5.81"	7.33"
240S	4"	---	1	12	5.82"	6.42"	8.20"

FEATURES

- Liquidtight, raintight, oiltight
- Suitable for wet locations
- Tapered threaded male hub NPT
- Insulated throat protects conductors from abrasion damage
- Compact slim diameter with small turning radius
- Zinc electro plated for excellent durability

APPLICATIONS

- To secure, ground and seal liquidtight conduit to a box or enclosure
- UL listed liquidtight flexible metal conduit fittings are suitable for use in the following hazardous locations under: NEC 501-4(b), 502-4(a), (2)503-3(a) Class 1 Div.2, Class II Div.1 and 2, Class III Div.1 and 2 and are suitable for grounding in sizes 3/8" through 1-1/2" under NEC 351-1

STANDARD MATERIAL

- Malleable iron
- Steel locknut

STANDARD FINISH

- Zinc plated

CERTIFICATIONS

- cULus

45° LIQUIDTIGHT CONNECTORS

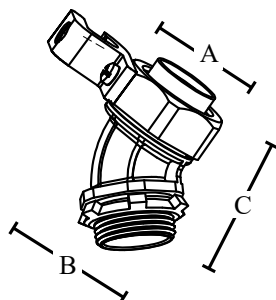
WITH GROUNDING LUG - MALLEABLE IRON



230SGR



230CSGR



FEATURES

- Liquidtight, raintight, oiltight
- Aluminum or Copper Lay-in grounding lug
- Insulated throat protects conductors from abrasion damage
- Zinc electro plated for excellent durability

APPLICATIONS

- Use to connect liquidtight conduit and terminate an external bonding conductor that runs parallel to the conduit
- The NEC code specifies length limitations and that the bonding jumper must be parallel to the conduit
- UL listed liquidtight flexible metal conduit fittings are suitable for use in the following hazardous locations under: NEC 501-4(b), 502-4(a), (2)503-3(a) Class 1 Div.2, Class II Div.1 and 2, Class III Div.1 and 2 and are suitable for grounding in sizes 3/8" through 1-1/2" under NEC 351-1

STANDARD MATERIAL

- Malleable iron
- Steel locknut
- Aluminum or Copper lug

STANDARD FINISH

- Zinc plated

CERTIFICATIONS

- UL 467 - UL 486A
- Fed. Spec: W-S-610E
- NEMA: GR 1-2017



PART #	SIZE	LUG SIZE	INNER QTY.	MASTER QTY.	WEIGHT/ CASE (LBS.)	A	B	C
ALUMINUM GROUNDING LUG								
230SGR	3/8"	14-4	5	50	16	1.18"	1.39"	2.51"
231SGR	1/2"	14-4	5	50	17	1.36"	1.50"	2.51"
232SGR	3/4"	14-4	2	50	24	1.61"	1.73"	2.70"
233SGR	1"	14-4	2	20	16	1.89"	2.11"	3.05"
234SGR	1-1/4"	14-4	2	20	24	2.39"	2.56"	3.65"
235SGR	1-1/2"	14-4	2	20	26	2.62"	2.77"	4.01"
236SGR	2"	14-4	2	20	37	3.24"	3.38"	4.64"
237SGR	2-1/2"	1/0-14	---	1	8	3.89"	4.68"	6.46"
238SGR	3"	1/0-14	---	1	11	4.74"	5.33"	7.01"
239SGR	3-1/2"	6/250	---	1	12	5.31"	5.81"	7.33"
240SGR	4"	6/250	---	1	14	5.82"	6.42"	8.20"
COPPER GROUNDING LUG								
231CSGR	1/2"	14-4	5	50	17	1.36"	1.50"	2.51"
232CSGR	3/4"	14-4	5	50	24	1.61"	1.73"	2.70"
233CSGR	1"	14-4	2	20	16	1.89"	2.11"	3.05"
234CSGR	1-1/4"	14-4	2	20	24	2.39"	2.56"	3.65"
235CSGR	1-1/2"	14-4	2	20	26	2.62"	2.77"	4.01"
236CSGR	2"	14-4	2	20	37	3.24"	3.38"	4.64"

LIQUIDTIGHT FITTINGS

Ocal® PVC-coated liquid-tight conduit connectors

The ultimate liquid-tight solution for corrosive environments



ST3/490-G 90° angled



ST3/4-G straight



ST3/445-G 45° angled

Product features

- Nominal 0.040" (40 mil) PVC coating bonded to exterior
- Standard color is dark gray – custom colors available upon request
- Pressure-sealing sleeves protect the connection
- ABB's Ocal brand uses only genuine T&B Liquidtight Systems® fittings to ensure quality installations
- **Ground ring meets UL 467 for grounding and bonding**
- Ground ring made with naval brass

Non-grounding connectors

PVC-coated steel straight product code	PVC-coated aluminum straight product code	PVC-coated steel 45° product code	PVC-coated steel 90° product code	PVC-coated aluminum 90° product code	Pipe size (in.)	Metric size designator*
ST3/8-	ST3/8SA-	ST3/845-	ST3/890-	ST3/890SA-	3/8	12
ST1/2-	ST1/2SA-	ST1/245-	ST1/290-	ST1/290SA-	1/2	16
ST3/4-	ST3/4SA-	ST3/445-	ST3/490-	ST3/490SA-	3/4	21
ST1-	ST1SA-	ST145-	ST190-	ST190SA-	1	27
ST11/4-	ST11/4SA-	ST11/445-	ST11/490-	ST11/490SA-	1 1/4	35
ST11/2-	ST11/2SA-	ST11/245-	ST11/290-	ST11/290SA-	1 1/2	41
ST2-	ST2SA-	ST245-	ST290-	ST290SA-	2	53
ST21/2-	ST21/2SA-	ST21/245-	ST21/290-		2 1/2	63
ST3-	ST3SA-	ST345-	ST390-		3	78
ST4-	ST4SA-	ST445-	ST490-		4	103
ST5-					5	129
ST6-					6	155

* Metric size designator (ANSI C80.1-1994).



ST3/490GR-G 90° angled



ST3/4GR-G-straight



ST3/445GR-G-45° angled

Connectors with grounding ring

PVC-coated steel straight product code	PVC-coated aluminum straight product code	PVC-coated steel 45° product code	Pipe size (in.)	Metric size designator*
ST1/2GR-	ST1/245GR-	ST1/290GR-	1/2	16
ST3/4GR-	ST3/445GR-	ST3/490GR-	3/4	21
ST1GR-	ST145GR-	ST190GR-	1	27
ST11/4GR-	ST11/445GR-	ST11/490GR-	1 1/4	35
ST11/2GR-	ST11/245GR-	ST11/290GR-	1 1/2	41
ST2GR-	ST245GR-	ST290GR-	2	53
ST21/2GR-	ST21/245GR-	ST21/290GR-	2 1/2	63
ST3GR-	ST345GR-	ST390GR-	3	78
ST4GR-	ST445GR-	ST490GR-	4	103

* Metric size designator (ANSI C80.1-1994).

Product Code	Color
ST3/4-	
	G = Dark gray
	W = White
	B = Light blue
GR in Product Code designates ground ring. Custom colors also available.	

Type NM SIM P U L NMLT

Non-Metallic Liquid-Tight Conduit. UL/CSA Listed, Crush, Oil, and Sunlight Resistant. Temperature Rated -30°C to 80°C.



Image not to scale. See Table 1 for dimensions.

CONSTRUCTION:

- PVC:** Type NM SIM Q V MMLT is manufactured with rigid PVC reinforcement imbedded within a flexible PVC wall. The construction provides excellent impact and crush strength while remaining highly flexible. The conduit resists oils, mild acids and exposure to sunlight.

2.03.C.2

APPLICATIONS AND FEATURES:

Type NM SIM Q V MMLT is suitable for the following installations:

- For the installation and protection of electrical conductors
- Motor circuits - for conductors of motor feeders, branch and control circuits.
- Where the conditions of installation, operation, or maintenance require flexibility or protection from liquids, vapors, solids, or weather
- Agricultural buildings per NEC® 547.5
- Used in industrial and commercial applications for conveyors, blowers, cranes, air conditioners, machine tooling and lubrication equipment
- Exposed or concealed installations
- Applications requiring movement, crossover connections, or tight bends
- For direct burial and encased in concrete
- For flexible connections to swimming pool, spa, and hot tub motors per NEC® 680.21(A)(3) & 680.42(A)(1)
- Electric signs and outline lighting supply and secondary-circuit conductors per NEC® 600.7, 600.31 (1000 volts or less) & 600.32 (over 1000 volts)
- Floating building feeders and services per NEC® 553.7
- Boatyards & Marinas in accordance with NEC® 555.13
- Cranes & Hoists in accordance with NEC® 610.11(C)
- For Elevator, Dumbwaiters, Escalators, Moving Walks, Wheel Chair Lifts & Stairway Chair Lifts in accordance with NEC® 620.21 (where expressly permitted)
- Under raised floors in information technology equipment conductors and cables in accordance with NEC® 645.5 (D) & 645.5(D)(2)
- Service entrance in lengths up to six feet per NEC® 230.43(15)
- For containment of 600 Volt and lower potential circuits
- Approved for use in lengths longer than 6 feet per NEC® 356.10(5)
- Motor leads per NEC® 430.145(B)

- Hazardous location:



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Southwire

**CABLETECH
SUPPORT™**

Services

UPDATED: March 11, 2024, 4:12 p.m. UTC REVISION: 1.000.039

- Class I, Div. 2 - NEC® 501.10(B)(2)
- Class II, Div. 1 - NEC® 502.10(A)(2)
- Class II, Div. 2 - NEC® 502.10(B)(2)
- Class III, Div. 1 - NEC® 503.10(A)(2)
- Class III, Div. 2 - NEC® 503.10(B)

Type NM SIM Q V MMT Features

- A protective combination of thermoplastic materials formed into a conduit/raceway which seals out water, liquids, abrasives, alcohol, coolants, corrosive fumes and gases, dirt, grease, mineral acids, non-concentrated fixed alkalines, petroleum oils, salt air and spray, and weather
- Smooth interior for easy wire pulling
- UV sunlight resistant jacket
- Rated for temperature range of -30°C to 80°C, 60°C Oil (-22 to +176, 140°F Oil)
- Accepts standard liquid-tight fittings for use with LFNC-B

2.03.C.2

2.03.C.4

2.03.C.2

2.03.C.1

SPECIFICATIONS:

- UL 1660 Liquid-Tight Flexible Nonmetallic Conduit
- CSA C22.2 No. 227.2.1 Liquid-Tight Flexible Non-Metallic Conduit
- NFPA 70 (National Electrical Code), Article 356 - Liquidtight flexible nonmetallic conduit

SAMPLE PRINT LEGEND:

SOUTHWIRE SIMpull™ LIQUID-TIGHT FLEXIBLE NONMETALLIC CONDUIT TYPE B X" (21) UL 80{D}C DRY 60{D}C WET 60{D}C OIL RES OUTDOOR DIR / BUR IN US SR --- UPC# 0-32886-89582-1

Type NM SOUTHWIRE® SIMpull™ NMLT

Trade Size	Inner Diam (Min/Max)	Outer Diam (Min/Max)	Weight	Minimum Bend Radius
Inch	Inch	Inch	Lb/100ft	Inch
1/2	0.622/0.642	0.820/0.840	129	6.5
3/4	0.820/0.840	1.030/1.050	178	8.5
1	1.041/1.066	1.290/1.315	259	13
1-1/4	1.380/1.410	1.630/1.660	309	16

Put Ups

TRADE SIZE (Inch)	50 ft	100 ft	250 ft	400 ft	1000 ft
1/2		13898401			13898402
3/4		13898501			13898502
1		13898601		13898602	
1 1/4	13898701		13898702		



2.0 EXPLOSION PROOF FLEXIBLE CONDUIT

SPECIFICATION – 16130–2.02.D FLEXIBLE CONDUIT



EXGJH and EXLK Flexible Couplings

Explosionproof, Dust-Ignitionproof

UNILETS™ for Use with Threaded Rigid Metal Conduit and IMC.

2.03.D.1

NEC/CEC — 1/2" and 3/4" Sizes:

Class I, Division 1 and 2,
Groups A, B, C, D
Class I, Zone 1 and 2, Groups IIA, IIB, IIC
Class II, Division 1 and 2, Groups E, F, G
Class III

NEC/CEC — 1" Size:

Class I, Division 1 and 2, Groups C, D
Class I, Zone 1 and 2, Groups IIA, IIB
Class II, Division 1 and 2, Groups E, F, G
Class III

NEC/CEC — 1-1/4" thru 4" Sizes:

Class I, Division 1 and 2, Group D
Class I, Zone 1 and 2, Group IIA
Class II, Division 1 and 2, Groups E, F, G
Class III

ATEX: — stainless steel 1/2" thru 2" Sizes:

Ex d IIC GD
Ex d IIC Gb
Ex t IIIC Db
IP66

Applications

- Used in areas where vibration and/or movement is a problem. Also used in place of rigid conduit in difficult-bend situations.

Features

- Explosionproof, dust-ignitionproof, watertight.
- Smooth, rounded integral bushing in each hub protects conductor insulation.
- Accurately tapped, tapered threads for tight, rigid joints and ground continuity.
- Heavy duty design resists mechanical abuse.
- Electrical conductivity equal to rigid conduit on a similar length basis—no bonding jumper required.
- Interior insulating liner protects conductors from abrasion under vibrating conditions.
- EXGJH: both end fittings are female, each furnished with a removable male nipple.
- EXLK: female end fitting with union at one end and a female end fitting with a removable male nipple at the other end.

Standard Materials

- 1/2" thru 2": outer bronze braid, inner brass core with insulating liner
- 2-1/2" thru 4": outer stainless steel braid, inner stainless steel core with insulating liner
- End Fittings: 1/2" thru 2" — brass; 2-1/2" thru 4" — stainless steel
- Unions: 1/2" thru 1" zinc plated steel; 1-1/4" thru 2" ductile iron
- Close nipples: zinc plated steel



EXGJH

Suitable for Wet Locations – Two Removal Male Nipples



EXLK

Suitable for Wet Locations – One Removal Male Nipple, One Removal Union

Standard Finishes

- Natural finish

Options

- 1/2" through 2" available in stainless steel. Add suffix **-SS**.
- EXLK Union is zinc plated steel. Stainless steel union available in 1/2" through 1". Add suffix **-SU**.
- Close nipples are zinc plated steel. For 1/2" through 1-1/2" stainless steel nipple. Add suffix **-NS**.
- Custom lengths available ① —consult factory.

NEC/CEC Certifications and Compliances

- UL Standard: 886 (1203)
- UL Listed: E34997 ②
- CSA Standard: C22.2 No. 25, C22.2 No. 30
- CSA Certified: 065181 ②

ATEX Certifications and Compliances

- Certification Type FLEXCO (*Excludes Nipples and Unions*)
 - Gas, Zones 1 and 2:
 - Conforming to ATEX 94/9/EC: Ex d IIC GD
 - Type of Protection: Ex d IIC Gb
 - Dusts, Zones 21 and 22:
 - Conforming to ATEX: Ex t IIIC Db
 - Type of Protection: Ex t IIIC Db
- Ambient Temperatures: -40 °C ≤ to ≤ + 150 °C (-40 °F ≤ to ≤ + 302 °F)
- Ex Standards: EN 60079-0, 60079-1, EN 60079-31
- ATEX Certificate: LCIE 11 ATEX 3006U
- Index of Protection according EN/IEC 60529: IP66

① Custom length flexible couplings greater than 36" are NOT UL listed.

② UL Listing and CSA Certification not applicable over 2" Trade Size.

EXGJH and EXLK Flexible Couplings

Explosionproof, Dust-Ignitionproof

UNILETS™ for Use with Threaded Rigid Metal Conduit and IMC.

NEC/CEC — 1/2" and 3/4" Sizes:

Class I, Division 1 and 2,
Groups A, B, C, D

Class I, Zone 1 and 2, Groups IIA, IIB, IIC
Class II, Division 1 and 2, Groups E, F, G
Class III

NEC/CEC — 1" Size:

Class I, Division 1 and 2, Groups C, D
Class I, Zone 1 and 2, Groups IIA, IIB
Class II, Division 1 and 2, Groups E, F, G
Class III

NEC/CEC — 1-1/4" thru 4" Sizes:

Class I, Division 1 and 2, Group D
Class I, Zone 1 and 2, Group IIA
Class II, Division 1 and 2, Groups E, F, G
Class III

ATEX: — stainless steel 1/2" thru 2" Sizes:

Ex d IIC GD
Ex d IIC Gb
Ex t IIC Db
IP66

Size (Inches)		Catalog Number	
Flex Length	Diameter Size	EXGJH	EXLK
4	1/2	EXGJH-14	EXLK-14
4	3/4	EXGJH-24	EXLK-24
6	1/2	EXGJH-16	EXLK-16
6	3/4	EXGJH-26	EXLK-26
6	1	EXGJH-36	EXLK-36
8	1/2	EXGJH-18	EXLK-18
8	3/4	EXGJH-28	EXLK-28
8	1	EXGJH-38	EXLK-38
10	1/2	EXGJH-110	EXLK-110
10	3/4	EXGJH-210	EXLK-210
10	1	EXGJH-310	EXLK-310
12	1/2	EXGJH-112	EXLK-112
12	3/4	EXGJH-212	EXLK-212
12	1	EXGJH-312	EXLK-312
12	1-1/4	EXGJH-412	EXLK-412
12	1-1/2	EXGJH-512	EXLK-512
12	2	EXGJH-612	EXLK-612
12	2-1/2	EXGJH-712	
12	3	EXGJH-812	
12	4	EXGJH-1012	
15	1/2	EXGJH-115	EXLK-115
15	3/4	EXGJH-215	EXLK-215
15	1	EXGJH-315	EXLK-315
15	1-1/4	EXGJH-415	EXLK-415
15	1-1/2	EXGJH-515	EXLK-515
15	2	EXGJH-615	EXLK-615
15	2-1/2	EXGJH-715	
15	3	EXGJH-815	
15	4	EXGJH-1015	
18	1/2	EXGJH-118	EXLK-118
18	3/4	EXGJH-218	EXLK-218
18	1	EXGJH-318	EXLK-318
18	1-1/4	EXGJH-418	EXLK-418
18	1-1/2	EXGJH-518	EXLK-518
18	2	EXGJH-618	EXLK-618
18	2-1/2	EXGJH-718	
18	3	EXGJH-818	
18	4	EXGJH-1018	
21	1/2	EXGJH-121	EXLK-121
21	3/4	EXGJH-221	EXLK-221
21	1	EXGJH-321	EXLK-321
21	1-1/4	EXGJH-421	EXLK-421
21	1-1/2	EXGJH-521	EXLK-521
21	2	EXGJH-621	EXLK-621
21	2-1/2	EXGJH-721	
21	3	EXGJH-821	
21	4	EXGJH-1021	

Size (Inches)		Catalog Number	
Flex Length	Diameter Size	EXGJH	EXLK
24	1/2	EXGJH-124	EXLK-124
24	3/4	EXGJH-224	EXLK-224
24	1	EXGJH-324	EXLK-324
24	1-1/4	EXGJH-424	EXLK-424
24	1-1/2	EXGJH-524	EXLK-524
24	2	EXGJH-624	EXLK-624
24	2-1/2	EXGJH-724	—
24	3	EXGJH-824	—
24	4	EXGJH-1024	—
27	1/2	EXGJH-127	EXLK-127
27	3/4	EXGJH-227	EXLK-227
27	1	EXGJH-327	EXLK-327
27	1-1/4	EXGJH-427	EXLK-427
27	1-1/2	EXGJH-527	EXLK-527
27	2	EXGJH-627	EXLK-627
27	2-1/2	EXGJH-727	
27	3	EXGJH-827	
27	4	EXGJH-1027	
30	1/2	EXGJH-130	EXLK-130
30	3/4	EXGJH-230	EXLK-230
30	1	EXGJH-330	EXLK-330
30	1-1/4	EXGJH-430	EXLK-430
30	1-1/2	EXGJH-530	EXLK-530
30	2	EXGJH-630	EXLK-630
30	2-1/2	EXGJH-730	
30	3	EXGJH-830	
30	4	EXGJH-1030	
33	1/2	EXGJH-133	EXLK-133
33	3/4	EXGJH-233	EXLK-233
33	1	EXGJH-333	EXLK-333
33	1-1/4	EXGJH-433	EXLK-433
33	1-1/2	EXGJH-533	EXLK-533
33	2	EXGJH-633	EXLK-633
33	2-1/2	EXGJH-733	
33	3	EXGJH-833	
33	4	EXGJH-1033	
36	1/2	EXGJH-136	EXLK-136
36	3/4	EXGJH-236	EXLK-236
36	1	EXGJH-336	EXLK-336
36	1-1/4	EXGJH-436	EXLK-436
36	1-1/2	EXGJH-536	EXLK-536
36	2	EXGJH-636	EXLK-636
36	2-1/2	EXGJH-736	
36	3	EXGJH-836	
36	4	EXGJH-1036	

◊ Indicates items in the shaded area which are UL Listed for Class I, Groups A, B, C, D; Class II, Groups E, F, G; Class III.

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EXGJH and EXLK Flexible Couplings

Explosionproof, Dust-Ignitionproof, Watertight

UNILETS™ for Use with Threaded Rigid Metal Conduit and IMC.

NEC/CEC — 1/2" and 3/4" Sizes:

Class I, Division 1 and 2,
Groups A, B, C, D
Class I, Zone 1 and 2, Groups IIA, IIB, IIC
Class II, Division 1 and 2, Groups E, F, G
Class III

NEC/CEC — 1" Size:

Class I, Division 1 and 2, Groups C, D
Class I, Zone 1 and 2, Groups IIA, IIB
Class II, Division 1 and 2, Groups E, F, G
Class III

NEC/CEC — 1-1/4" thru 4" Sizes:

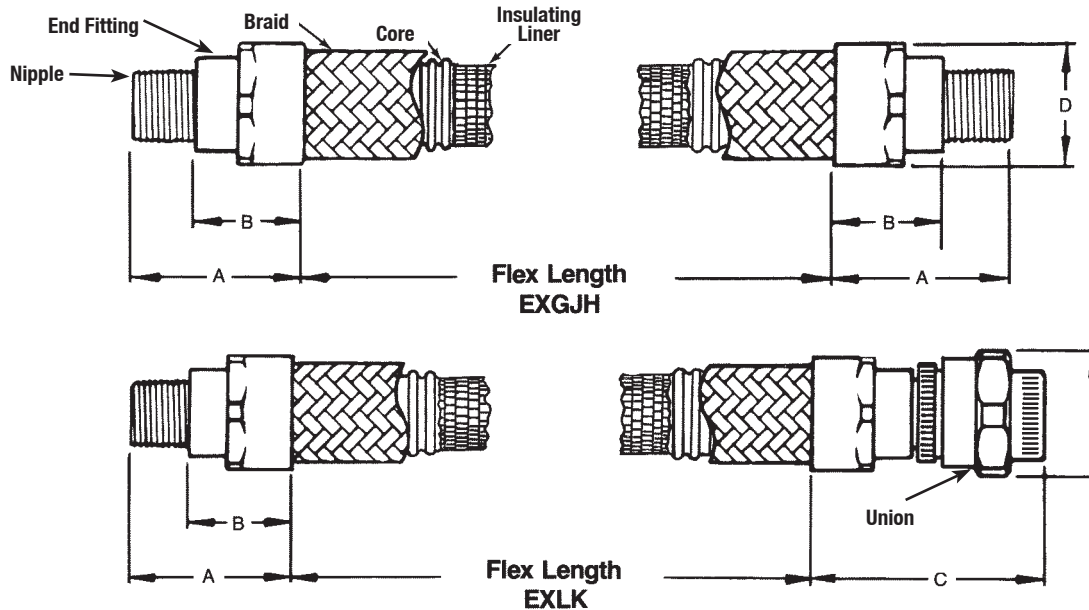
Class I, Division 1 and 2, Group D
Class I, Zone 1 and 2, Group IIA
Class II, Division 1 and 2, Groups E, F, G
Class III

ATEX: — stainless steel 1/2" thru 2" Sizes:

Ex d IIC GD
Ex d IIC Gb
Ex t IIC Db
IP66

EXGJH and EXLK Dimensions in Millimeters (Inches)

Coupling Size	A	B	C	D	E	Bending Radius ①
1/2	80.7 (3.18)	38.1 (1.50)	76.2 (3.00)	35.7 (1.41)	37.3 (1.47)	254.0 (10.00)
3/4	81.0 (3.19)	38.9 (1.53)	82.6 (3.25)	44.5 (1.75)	44.5 (1.75)	304.8 (12.00)
1	91.9 (3.62)	46.7 (1.84)	92.1 (3.63)	52.4 (2.06)	50.8 (2.00)	355.6 (14.00)
1-1/4	98.0 (3.86)	51.6 (2.03)	111.1 (4.38)	66.7 (2.63)	71.4 (2.81)	406.4 (16.00)
1-1/2	108.5 (4.27)	55.6 (2.19)	115.9 (4.56)	79.4 (3.13)	77.8 (3.06)	406.4 (16.00)
2	114.3 (4.50)	62.0 (2.44)	125.4 (4.94)	98.4 (3.88)	94.5 (3.72)	457.2 (18.00)
2-1/2	133.4 (5.25)	65.0 (2.56)	158.8 (6.25)	114.3 (4.50)	123.8 (4.88)	381.0 (15.00)
3	136.1 (5.36)	66.8 (2.63)	158.8 (6.25)	127.8 (5.03)	136.5 (5.38)	584.2 (23.00)
4	140.7 (5.54)	66.8 (2.63)	161.9 (6.38)	168.3 (6.63)	163.5 (6.44)	1219.2 (48.00)



① Bending Radius shown is minimum recommended.